AN INTERACTIVE RESTAURANT MENU

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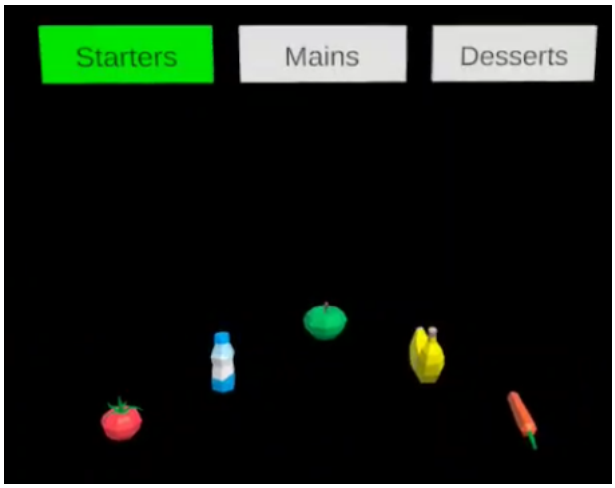

In an era where augmented reality is becoming more and more integrated into our daily lives, this paper explores the potential of AR to enhance the dining experience through an innovative approach at desining a restaurant menu. We explore an AR 3D restaurant menu displayed in a carousel format, which diners interact with using AR glasses. The menu is operated through a touchpad and enhanced with head-gaze technology. Our menu aims to replace traditional paper menus by replacing them with a more interactive and immersive process, in which the user gets to visualize their food before ordering. The evaluation yielded mixed but generally positive feedback, indicating strong potential. Key findings suggest that even with an initial learning curve, the intuitive nature of the menu promises a fast and intuitive alternative to traditional menus. Future work will focus on minimising initial complexity, while also expanding on missing features that further enhance the user experience.

Additional Key Words and Phrases: AR, VR, Restaurant, Menu, Carousel Menu, Touchpad

1 INTRODUCTION

As virtual reality (VR) and augmented reality (AR) technologies become increasingly integrated into daily life, the need to develop applications that enhance user experience without complicating functionality is paramount. This paper presents an application of augmented reality (AR) technology that aims to revolutionize the traditional restaurant menu experience. Using AR, our project aims to transform the way users interact with menus in a restaurant setting, offering a more intuitive and immersive selection process.

The concept involves designing a digital menu that users can navigate through a touchpad located on their hand. This menu is a dynamic display where users can view realistic digital representations of food items, including actual portion sizes, before making a selection. This system aims to mitigate the uncertainty often associated with ordering food in restaurants and to enhance customer satisfaction by providing a more informed dining decision. Once an order has been placed, the menu is simply put away to not disrupt the restaurant dining experience we know today.

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Our approach addresses the challenges associated with incorporating AR into everyday environments, particularly focusing on user interaction paradigms that do not disrupt the conventional dining experience. The design process, which evolved through multiple iterations, experimented with both two-dimensional and three-dimensional interface models to determine the most effective method for menu interaction in AR.

In the end we developed a restaurant menu built up of three components, of which we will go into further detail later in the report, which utilises a carousel menu as its cornerstone with auxiliary buttons for further navigation. The carousel is navigated with the aforementioned touchpad which is located on the user's hand.

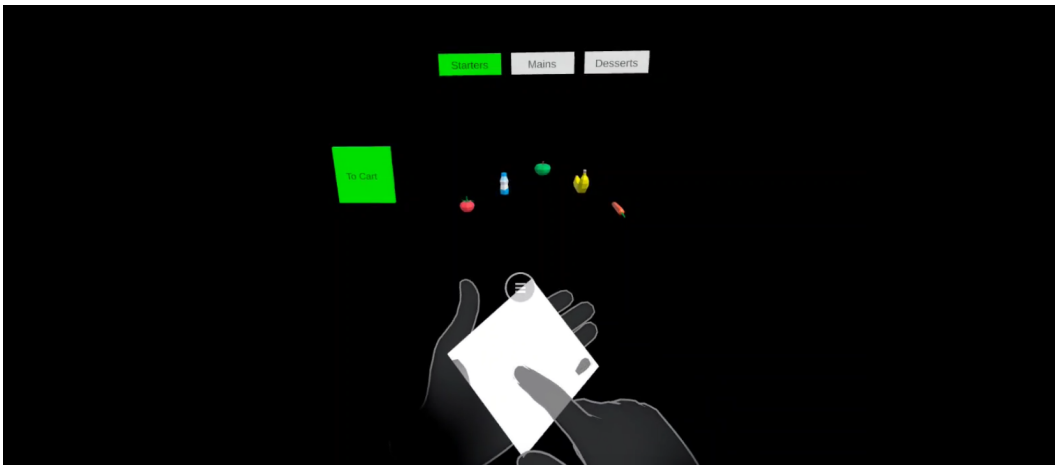

Fig. 1. A screenshot of our end prototype, showcasing the caorusel menu and touchpad on the users hand as well as auxiliary buttons to switch between starters, mains etc. as well as the button that switches between the cart and the menu.

A Video demonstration of the menu can be found here on YouTube: https://youtu.be/xnTdcsNSfME

2 BACKGROUND

We have found that others have also explored the notion of an interactive restaurant menu. The first design we looked at is a design called Mojo iCuisine. It is not an AI solution, but rather an "immersive" solution that utilises both a projector and a touch-enabled table. The design is an actual business application and is "explorative and service-oriented" [2]. This is a central point for us, as we also approached our design process from an exploratory angle.

Three-dimensional menus: A survey and taxonomy by Dachselt and Hübner suggests that we should move away from the traditional 2D menu types, that is, acyclic lists. Currently, despite an additional dimension, a lot of 3D menus are still lists. As an alternative, they suggest exploring other geometric structures, which inspired us to explore the option of a carousel menu. The inspiration for our carousel menu comes from the so-called ring menu mentioned in the paper, in which a list of items is displayed in a ring, which the user rotates with their hand. [3]

3 CONCEPT(S)

In the development of the architectural framework for the menu prototype, multiple design alternatives were explored. Initially, the design of the prototype closely mirrored conventional real-world menus or those typically encountered in smartphone applications. The idea was that a touch plane

would be attached to the user's hand, which could then be used to navigate the menu displayed on another flat plane. The user would then double-tap on the touch plane to select the desired item, which would then be added to the cart. As well as getting added to the cart, the item would also be displayed as a 3D model on the table on the users table. This would allow the user to see the food before ordering, as well as get an idea about the portion sizes. We also considered adding a timer that displays the estimated time of the food's arrival. This initial design iteration was ultimately deemed too 2D and insufficiently innovative for the capabilities offered by a 3D interface environment. Suggestions of refinements led to the second prototype iteration, which

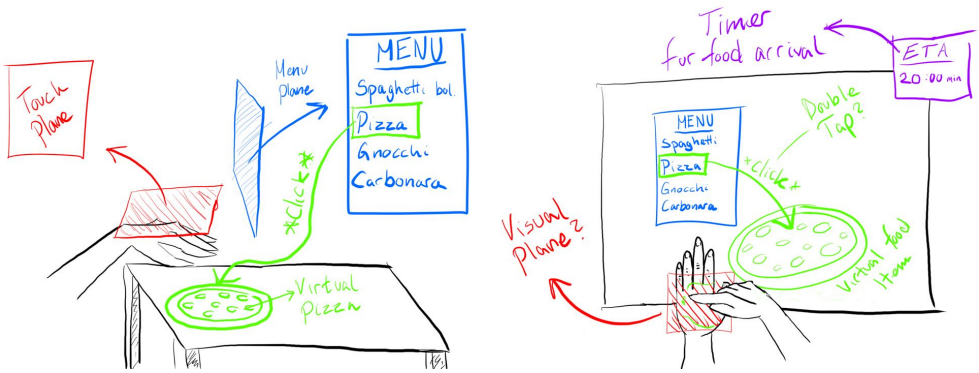

Fig. 2. 1. Iteration design of menu prototype seen from two angles - A touchpad is attached to the users hand, which is used to navigate the 2D menu resembling that of a normal restaurant menu. Upon selection of an item (In this case a pizza) it is displayed on the table.

utilises the 3D space more effectively. The revised menu configuration presents itself as a carousel, positioned in front of the user, who can navigate through the selections to identify and choose the preferred item. This idea as mentioned in a previous chapter is derived from the ring menu as described in the paper *Three-dimensional menus: A survey and taxonomy* by Dachselt and Hübner. [3] The user navigates the menu in the same way as before by using the touchpad attached to the user's hand, and selection is still done by double-tapping the touchplane. Above the carousel, a categorical navigation bar lists various types of food: snacks, appetizers, mains, and desserts. Interaction with this bar enables users to navigate to specific food categories quickly. In addition, a peripheral button switches the display to a shopping cart, also formatted as a carousel. Within this cart, users can modify the quantity of items or remove them entirely.

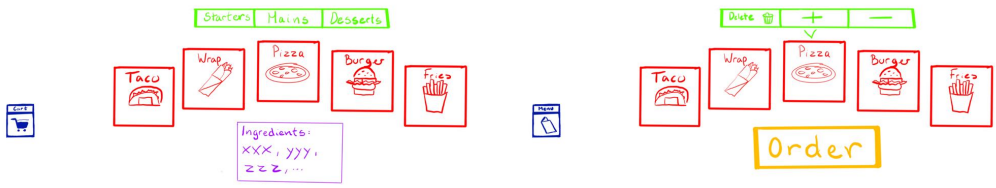

Fig. 3. 2.1. Iteration design of menu prototype - Left side shows the carousel menu displaying foods that can be selected. The user can swipe left or right to cycle through different food items. The right picture displays the view from the cart, which shows items already added to the cart. The user can switch between the menu and the cart using the button on the left.

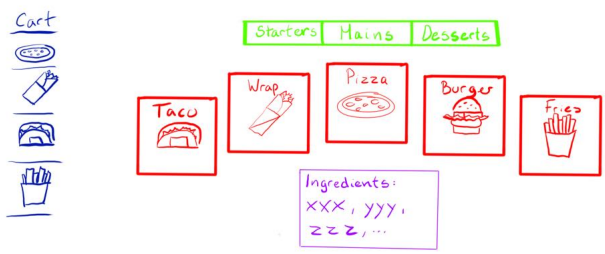

Fig. 4. 2.2. Iteration design of menu prototype - This image showcases our second idea for displaying the cart, in which the items added to cart are displayed in a stack to the left of the carousel. The user can look at the cart and use the touchpad to cycle through the items as with the items in the menu.

In the second prototype iteration, we had different ideas for implementing the peripheral button that switches between the menu and the cart. One of the ideas was to implement a simple button with a cart icon, and the way you could see if you had added something to the cart, would be a numerical indicator, like the ones we are used to seeing above smartphone apps. We liked that idea because of its simplicity and how it would align with what the user is used to seeing on their phone and computer. Ultimately, we decided to adopt this design for our implementation.
Another design option we considered for the cart button was to implement a vertical carousel. In this implementation, the user wouldn't have to switch between the menu and the cart; instead, they would use the horizontal carousel to select food items and the vertical carousel to edit the cart. The vertical cart would consist of miniature versions of the added food items, and users could

switch to the cart using gaze detection. To edit an item, the user would double-tap the miniature item they wanted to edit, which would display a small menu allowing them to add, subtract, and delete the food item.

In the end, we thought that adding another carousel with a different orientation would ultimately confuse the user. Therefore, we opted to implement the solution we believed would be the simplest for the user to learn.

4 IMPLEMENTATION

For full implementation check GitLab [1]

4.1 Hardware

Our prototype is designed for the featureset of the Meta Quest 2, however other VR headsets with a similar featureset should suffice. What we required from the hardware was passthrough, such that you could still interact with the real world around you while looking at the menu, creating a more immersive experience. Not having passthrough would take away from the traditional experience of sitting in a restaurant with a menu. We also required the headset to have hand tracking for use with the touchpad, as well as head-gaze such that the menu knows which part you are interacting with. Eye-gaze would have been preferable, as it feels more natural to simply move your eyes than having to turn your head when wanting to interact with different parts of the menu, however we found head-gaze to be sufficient for our needs.

4.2 Software

4.2.1 The Touchpad. The touchpad serves as the cornerstone of interaction for our application. Due to the significance of the touchpad within our project we wanted it to have certain properties. We wanted it to be intuitive and always within reach. We therefore opted to attach the touchpad to the users hand. We considered and tested a couple of options for the placement of the touchpad, and found the palm to feel most natural.

Upon interacting with the touchpad, the initial coordinate of where the index finger touches the touchpad is recorded, and all subsequent coordinates thereafter until the user removes their finger from the touchpad. We then measure the difference in the initial x-coordinate and the end x-coordinate and measure this up against a threshold. If the difference is larger than the threshold we consider the interaction to be a swipe. As the touchpad is attached to the users hand, their swipe is initially registered in relation to the global space. This meant that should the user turn your hand, they would no longer be swiping on the x-axis. To accurately interpret the swipes we transformed the global coordinates to the local coordinate system of the touchpad, through so-called "InverseTransformPoint", see Figure 5. To further interact with the touchpad, the user can also tap it to perform a click-action known from laptop touchpads. We initially struggled with accurate swipes and taps due to the subpar finger tracking of the Quest 2, however the threshold allowed an adjustable margin of error.

4.2.2 The Menu / Head-Gaze. While we won't be going into full detail on how the menu is implemented, it is important to highlight the role that Head-Gaze plays in the overall experience of our menu. We utilize Head-Gaze to seamlessly transition between the three sections of the menu: the main carousel menu, the sub-menu above the carousel, and the button that switches between the cart and the store, see Figure 6. With the touchpad's functionalities limited to swipe and tap, Head-Gaze allows the user to effortlessly navigate between these sections by simply directing their gaze. All of this is implemented using raycast, which interacts with the colliders of the subsections of the menu.

```
18      private void OnTriggerStay(Collider other)
19      {
20          const float threshold = 3f;
21          var diff = transform.InverseTransformPoint(other.transform.position) - pos;
22          xDiff += diff.x;
23          if (xDiff > threshold)
24          {
25              hasSwiped = true;
26              if (menuScript.getActiveMenu() == "FoodSubMenu")
27              {
28                  menuScript.IncrementFocus();
29              }
30              xDiff = 0;
31          }
32          else if (xDiff < -threshold)
33          {
34              hasSwiped = true;
35              if (menuScript.getActiveMenu() == "FoodSubMenu")
36              {
37                  menuScript.DecrementFocus();
38              }
39              xDiff = 0;
40          }
41          pos = transform.InverseTransformPoint(other.transform.position);
42      }
```

Fig. 5. Some of the most important code of our touchpad script. This involves everything that happens while ones finger interacts with the collider of the touchpad. The difference in coordinates fromt the last frame is calculated on line 21, as well as the position getting reset at line 41. There we see the use of the InverseTransformPoint to transform from global to local coordinates. In line 23 we see the if statement for the threshold of when something is considered a swipe.

```
void Update()
{
    RaycastHit hit;
    var transform1 = cameraRig.transform;
    var forwardDirection = cameraRig.centerEyeAnchor.TransformDirection(Vector3.forward);
    if (Physics.Raycast(cameraRig.centerEyeAnchor.position, forwardDirection, out hit)) {
        activeMenu = hit.transform.gameObject;
    }
```

Fig. 6. This figure showcases the simple headgaze logic that gets the gameObject of the collider that the user is currently looking at. This allows them to change which menu the touchpad interacts with simply by moving their head.

5 EVALUATION

We conducted our evaluation using a System Usability Scale (SUS). We opted to use SUS due to it being a quick, and reliable method of assessing user satisfaction and gathering feedback regarding usability. During the evaluation, we would give the participants tasks to complete, such as "navigate to the dessert menu", "add a donut to the cart", "remove the donut from the cart" etc. If participants struggled or got stuck, we would hin them towards the solution without telling them exactly what to do. We used a standard SUS, but modified it slightly by adding five more questions that were more directed at our prototype, more specifically aimed at getting feedback for the touchpad. The SUS consisted of two sections, the first section includes 12 questions answered on a Likert scale, and the second section includes open-ended feedback. Two of the five questions we added were added to the Likert scale section. For the Likert scale section results, see Figure 8.

Based on our questionnaire we calculated the SUS score, which is a measurement from 0 to 100 on how usable your system is, where a higher score means a more usable system. Normally a SUS questionnaire has 10 questions scored 1 to 5, we however added 2 more. We adjusted for this in our SUS score by normalizing the results. We recorded a max SUS score of 91.67, a mimimum SUS score of 41.67, and an average SUS score of 60.42. For all SUS scores see Figure 7.

Participant	**1**	**2**	**3**	**4**	**5**
SUS Score	47.92	41.67	91.67	75	45.83
Average	60.42				

Fig. 7. SUS scores

In the open-ended feedback section we added three questions, which are not a part of a normal SUS, aiming to get more direct feedback about the interaction with the touchpad. For the results of the open-ended feedback see Figure 9

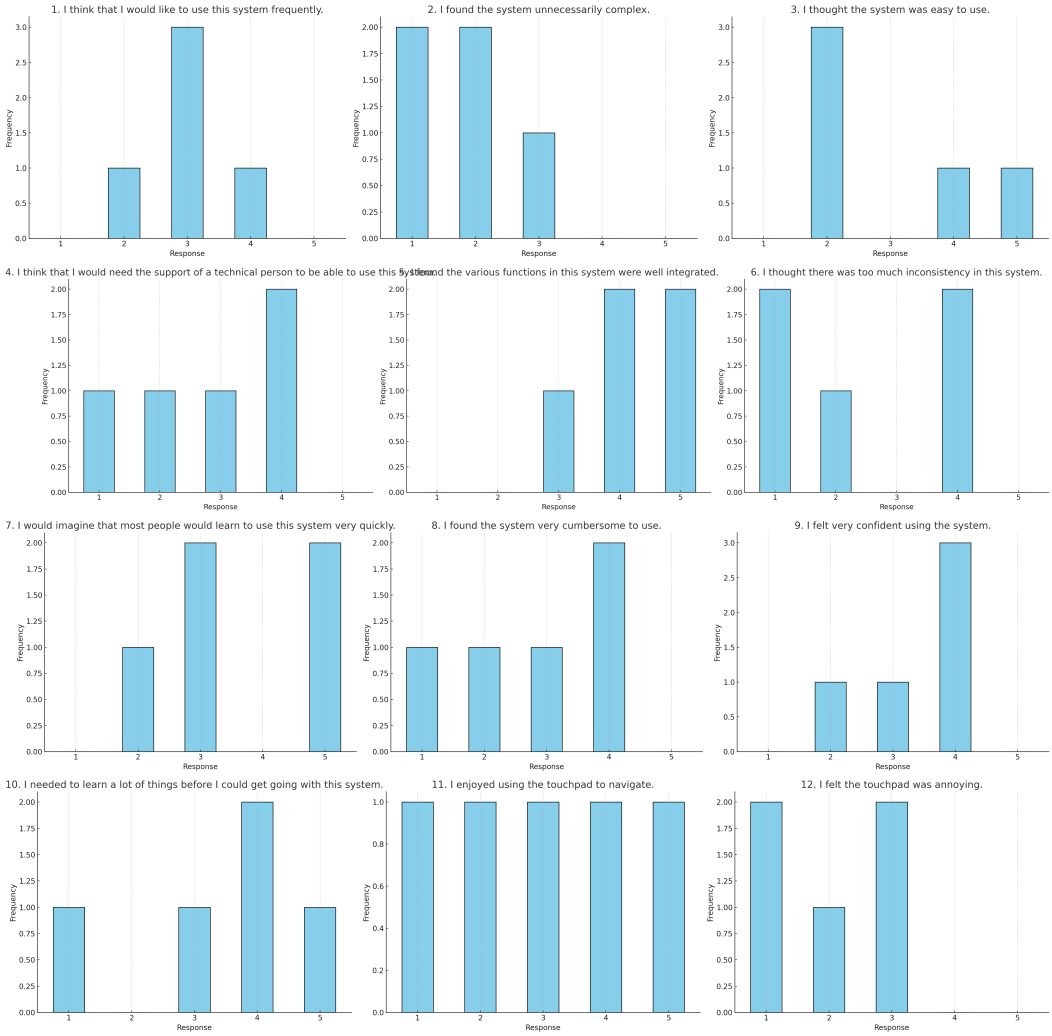

Fig. 8. Likert scale questions for the SUS questionaire

Person	What did you like about the solution you just experienced?	What did you dislike about the solution you just experienced?
1	I liked the idea of navigating between items with ease.	Bugginess of actually triggering things going forward.
2	Once I learned how to use and navigate the solution, it was intuitive.	Coordination was off and I had to adjust frequently.
3	It was easy to shop, after I figured out the system.	The room initialization set of position was a bit buggy.
4	It is quite simple to use. Scrolling worked nicely.	Confirmation of selection (tap) didn't work for some items.
5	Fun way to do menu	Difficult to use in the beginning

Person	What did you like about using the touchpad to navigate?	What did you dislike about using the touchpad to navigate?
1	It is similar to using something like a smartphone.	It is just a bit too buggy so it was super hard to navigate.
2	I feel like the touchpad is not necessary.	Just a white box which I thought didn't describe its function well.
3	It was super easy to understand.	The head ray and room initialization made it a bit hard.
4	Speed.	Inertia wasn't implemented.
5	I didn't have to look at it when I got the hang of it.	I wanted to look at it all the time.

Person	Would you like to change anything about how the touchpad works? If so, what?
1	Better feedback and tracking of what is going on.
2	I would remove it and use just hands and hand gestures.
3	Just the initialization in the room space.
4	Nothing. It's awesome.
5	Make it see-through so I don't look at it all the time.

Fig. 9. Open-ended responses for the SUS questionnaire

6 DISCUSSION

In this chapter we will interpret the results of our evaluation of our prototype. As mentioned in the previous chapter, we conducted the evaluation using a System Usability Scale (SUS), to quantitaviely measure the usability of our prototype. The SUS had an average score of 60.42 as seen in Figure 7, which indicates that there are problems with the usability of our system, as it is slightly below the average SUS core of 68. [4] This indicates that our prototype has room for improvement, which contradicts our hypothesis that our system would be intuitive and easy to use for beginners. To see why this may be we look to the open-ended part of our evaluation which had some recurring themes, such as the common opinion that our system has a high ease of use once one is already accustomed to how the system functions. There were also some common criticisms regarding the initial complexity of the system. We were under the impression that most people would find the touchpad intuitive, as we modeled it after touchpads known from modern laptops. However, during the study, we noticed that people kept glancing at the touchpad while trying to interact with the system, which due to the nature of how our system utilises head-gaze caused issues with navigation. This indicates that perhaps to some users, the metaphor of the touchpad does not translate as well as we had wished for. We did also receive very positive feedback to the touchpad, specifically with participants 3 and 4 as seen in Figure 9. They had few to no issues navigating through the system, and found the touchpad combined with head-gaze to be fast and intuitive.

During the evaluation we also received some recommendations on how to improve the prototype, such as adding inertia. This would allow the user to scroll faster depending on how fast they swipe their finger across the touchpad. A user also recommended making the touchpad see-through, as this would make them less likely to look at it, thus minimising errors.

Overall we received very balanced feedback with some users really liking the system and others disliking it.

7 CONCLUSION AND FUTURE WORK

In this project, we set out to develop an intuitive and modern AR solution to traditional restaurant menus. We accomplished this task by developing a restaurant menu in form of a carousel menu which is operated through a touchpad in cooperation with head-gaze. Our evaluation revealed that while our system has a learning curve, it becomes intuitive and efficient once accustomed to it. Although feedback from our evaluation was mixed, we did receive a lot of positive feedback, and believe there is potential for improving upon the system. Further research should focus on minimising initial complexity, as well as implementing missing features such as inertia, which would increase efficiency. Another feature for future development is displaying the selected items on the user's table as discussed in previous chapters.

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